

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

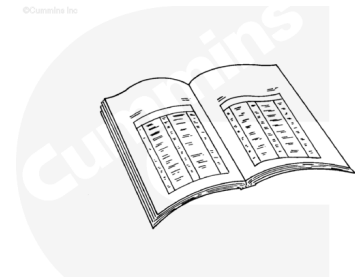
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

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- Drain the lubricating oil from the engine. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/441/441-007-037.html)
- Drain the coolant from the engine . Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/441/441-008-018-tr.html)
- Remove the engine from the chassis. Refer to Procedure 000-001 in Section 0.
(/qs3/pubsys2/xml/en/procedures/441/441-000-001.html)
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/441/441-003-009.html)
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/441/441-004-014.html)
- Remove the front gear cover up to the gear cover plate. Refer to Procedure 001-031 in Section 1.
(/qs3/pubsys2/xml/en/procedures/441/441-001-031.html)

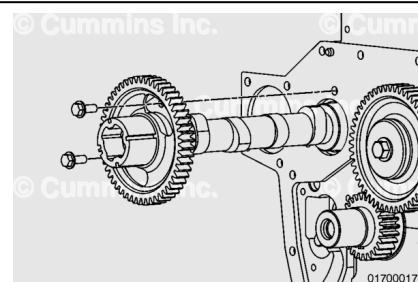
Remove

Position the engine so the crankshaft is above the camshaft. This will keep the tappets from falling during camshaft removal.

Remove the two mounting capscrews through the casting holes in the camshaft gear.

Note : Rotate the camshaft slightly during removal. Use care to **not** damage the camshaft bushings and machined surfaces.

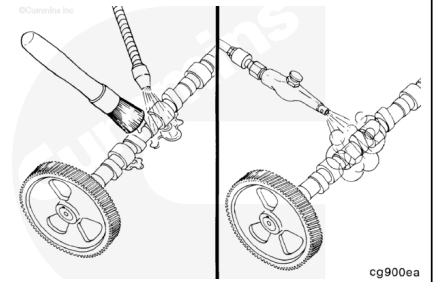
Remove the camshaft.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

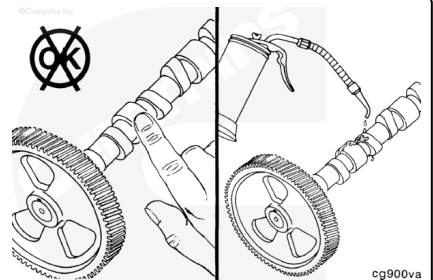
Use solvent or steam to clean the camshaft.

Dry with compressed air.

⚠ CAUTION ⚠

Do not touch the machined surfaces with bare hands; this can cause rust to form on the camshaft.

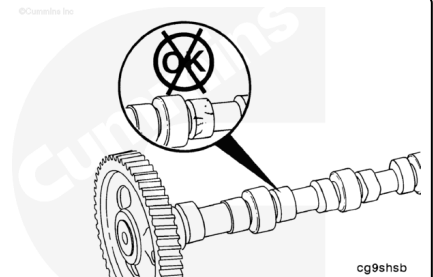
Before the camshaft is touched, lubricate the camshaft with clean engine lubricating oil.



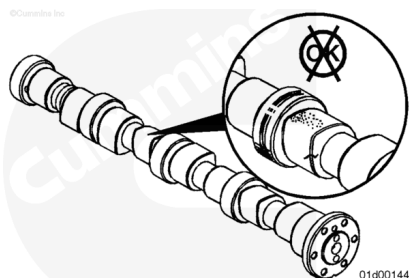
Note : When a new camshaft is installed; new tappets and push tubes must be installed also.

Inspect the bearing journals for cracking, pitting, scoring, or other damage.

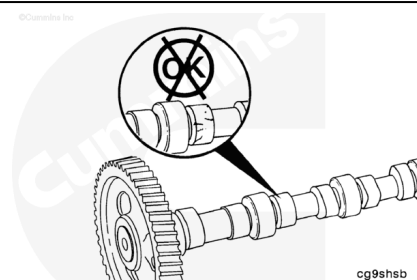
Contact a Cummins® Authorized Repair Location for pitting reuse criteria or use the following reuse guidelines.



Inspect the valve lobes and bearing journals for cracking, pitting, scoring, and other damage.

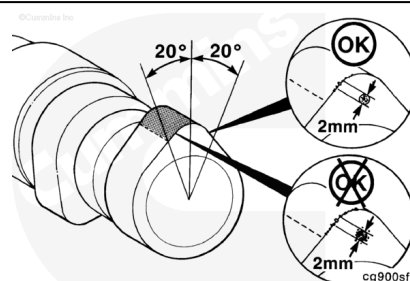


For guidelines for reuse of cast iron camshaft and tappets, use the Camshaft and Tappet Reuse Guidelines, Bulletin 3666475. (</qs3/pubsys2/xml/en/bulletin/3666475.html>)



Edge Deterioration (breakdown) Criteria

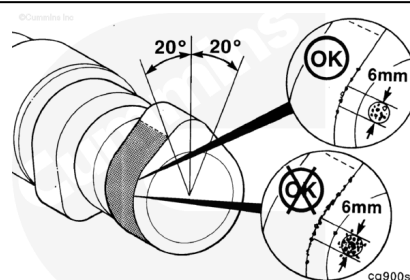
The area of edge deterioration **must not** be greater than the equivalent area of a 2 mm [0.079 in] circle within ± 20 degrees of the nose of the camshaft lobe.



Outside of the ± 20 degrees of the nose of the camshaft lobe, the areas of edge deterioration **must not** be greater than the equivalent area of a 6 mm [0.236 in] circle.

Note : If the camshaft shows any pitting or wear, remove and inspect the tappets before installing the camshaft. Refer to Procedure 004-015 in Section 4. (</qs3/pubsys2/xml/en/procedures/441/441-004-015.html>) If a new camshaft is installed, new tappets **must** be installed.

Note : When a new camshaft is installed on engines with more than 250 hours, 5000 miles, or 8000 kilometers of engine operation, new tappets and push rods **must** also be installed.

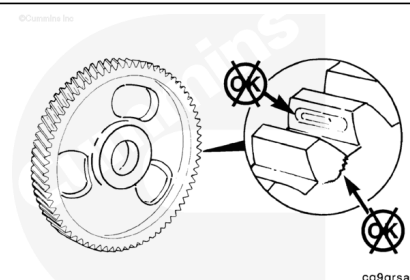


Inspect the camshaft gear teeth for pitting.

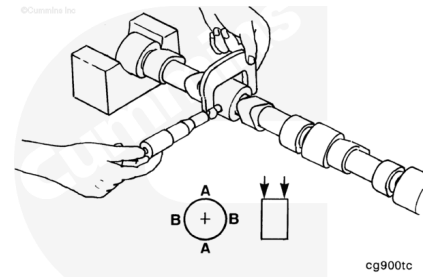
Inspect for cracks at the roots of the teeth.

Replace the camshaft gear if damage is found. Refer to Procedure 001-013 in Section 1.

(</qs3/pubsys2/xml/en/procedures/441/441-001-013.html>)



Measure the outside diameter of the camshaft bearing journals.



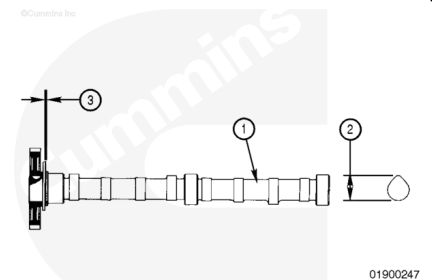
Camshaft Bearing Journals

Journal	Standard Size	Tolerance		Standard Clearance	Clearance Limit
		Shaft	Hole		
Number 1, Number 2, and Number 3	50.5 mm [2.00 in]	-0.080 to -0.110 mm [-0.003 to -0.004 in]	+0.030 to -0.040 mm [+0.001 to -0.002 in]	0.040 to 0.140 mm [0.006 to 0.016 in]	0.25 mm [0.010 in]

Measure the bend (1) of the camshaft.

Bend of Camshaft

mm		in
0.03	MAX	0.0012



Measure the camshaft lobe height (2).

Camshaft Lobe Height

	mm		in
Intake Valve	41.81440	NOM	1.646231
Exhaust Valve	42.40479	NOM	1.669475

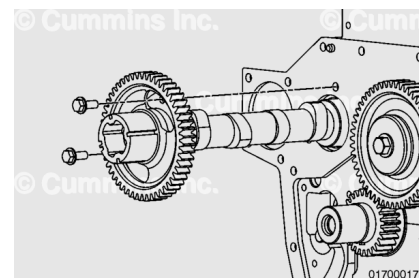
Measure the camshaft end clearance (3).

Camshaft End Clearance

mm		in
0.150 to 0.355	NOM	0.0059 to 0.0138
0.35	MAX	0.013

Install

Install the camshaft slowly. While pushing lightly, rotate the camshaft to carefully work the camshaft through the camshaft bushings. As each camshaft journal passes through a bushing, the camshaft will drop slightly and the camshaft lobes will catch on the bushings. Rotating the camshaft will free the lobe from the bushing and allow the camshaft to be installed.



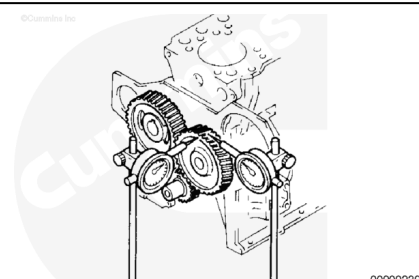
Note : When installing the camshaft, rotate the camshaft slightly and use care to not damage the camshaft bushings and machined surfaces.

Install the thrust plate and two capscrews.

Tighten the capscrews through the casting holes in the camshaft gear.

Torque Value: 31 n•m [23 ft-lb]

Measure the end clearance of the camshaft. Use a gauge, Part Number 3824564, and a magnetic base, Part Number 3377399, to check the camshaft end clearance.

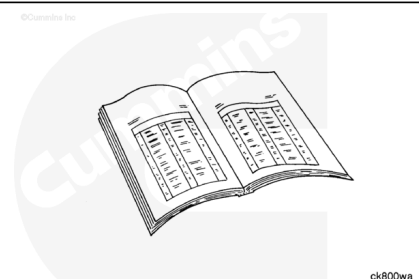


Camshaft End Clearance		
mm		in
0.150	MIN	0.0059
0.350	MAX	0.0138

Finishing Steps

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Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

** WARNING **

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- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/441/441-001-031.html>)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/441/441-004-014.html>)
- Install the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/441/441-003-009.html>).
- Install the engine in chassis. Refer to Procedure 000-002 in Section 0. (</qs3/pubsys2/xml/en/procedures/441/441-000-002.html>)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/441/441-008-018-tr.html>)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/441/441-007-037.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.